Binomial Coefficients and the Structure of an Expansion Answer Key

Algebra 2

Quick Answers

1. $x^3 + 3x^2 + 3x + 1$
2. $x^3 - 12x^2 + 48x - 64$
3. 40
4. 625
5. 32
6. $16x^4 - 32x^3 + 24x^2 - 8x + 1$
7. -720
8. (a) = 84, —
9. 63
10. 15
11. $x^6 + 9x^4 + 27x^2 + 27$
12. (a) = $x^4 + 4x^3 + 6x^2 + 4x + 1$, (b) = 81, (c) the exponent = 9, —

What is verified

14 of 16 answers machine-checked against SymPy, across 12 problems.
— marks an answer only you can judge — problems 8, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 2
HSA-APR.C.5 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 10, 11, 12*
HSF-BF.A.2 – *problem 9*
Difficulty 1–5 of 5 across 16 tagged checks.

1. Expand $(x + 1)^3$.

Solution: Row 3 of Pascal's triangle is $1, 3, 3, 1$, and every power of 1 is 1, so the coefficients are the row itself. $\boxed{x^3 + 3x^2 + 3x + 1}$

2. Expand $(x - 4)^3$.

Solution: Same row $1, 3, 3, 1$, now with $B = -4$, so the powers $(-4)^1$, $(-4)^2$, $(-4)^3$ alternate in sign.

$$(x - 4)^3 = x^3 + 3x^2(-4) + 3x(16) + (-64)$$

$$\boxed{x^3 - 12x^2 + 48x - 64}$$

3. Coefficient of x^3 in $(x+2)^5$.

Solution: With $n=5$, the x^3 term needs x three times and 2 twice, so $k=2$.

$$\binom{5}{2}x^3 2^2 = 10 \cdot 4 \cdot x^3$$

The coefficient is $\boxed{40}$

---

4. Sum of all coefficients of $(2x+3)^4$.

Solution: Substituting $x=1$ turns every power of x into 1, so each term collapses to its own coefficient and the whole expression becomes the coefficient total.

$$(2 \cdot 1 + 3)^4 = 5^4$$

$\boxed{625}$

---

5. Evaluate $\sum_{k=0}^{5} \binom{5}{k}$.

Solution: The sum runs over the whole of row 5: $1, 5, 10, 10, 5, 1$.

$$1 + 5 + 10 + 10 + 5 + 1 = 32$$

$\boxed{32}$

This is 2^5, and that is not a coincidence: it is problem 4's substitution with $A = B = 1$.

---

6. Expand $(2x-1)^4$.

Solution: Row 4 is $1, 4, 6, 4, 1$, with $A = 2x$ (so each power of A squares, cubes, or fourth-powers the 2 as well) and $B = -1$.

$$(2x-1)^4 = 16x^4 + 4(8x^3)(-1) + 6(4x^2)(1) + 4(2x)(-1) + 1$$

$\boxed{16x^4 - 32x^3 + 24x^2 - 8x + 1}$

---

7. Coefficient of x^2 in $(3x-2)^5$.

Solution: The x^2 term takes $3x$ twice and -2 three times, so $k=3$.

$$\binom{5}{3}(3x)^2(-2)^3 = 10 \cdot 9x^2 \cdot (-8)$$

The whole factor $3x$ is squared, and $(-2)^3$ keeps its sign. The coefficient is $\boxed{-720}$

---

8. (a) Compute $\binom{9}{3}$. (b) Why does it equal $\binom{9}{6}$?

Solution (a):

$$\binom{9}{3} = \frac{9!}{3!\,6!} = \frac{9 \cdot 8 \cdot 7}{6} = \frac{504}{6}$$

$\boxed{84}$

Solution (b) — instructor-judged. Choosing which 3 of the 9 factors contribute their second term is the same act as choosing which 6 do not, so the two counts must be equal. Written out, both are 9! over 3! 6! — the same denominator with its two factors in the other order. In the expansion this is the left-right mirror symmetry of a row of Pascal's triangle. Full credit gives a reason of that kind. Computing $\binom{9}{6} = 84$ and noting the match is a check, not an explanation — partial credit.

---

9. Evaluate $\sum_{n=0}^{5} 2^n$.
Solution: This is a geometric series with first term 1 and ratio 2, six terms long.

$$1 + 2 + 4 + 8 + 16 + 32 = \frac{2^6 - 1}{2 - 1}$$

$\boxed{63}$

Each term is one row's coefficient total, so this is the number of entries' worth of weight in the first six rows of Pascal's triangle taken together.

---

10. Coefficient of x^4y^2 in $(x + y)^6$.
Solution: Two letters, no numerical factors, so the coefficient is the binomial coefficient alone. The y appears twice, so $k = 2$:

$$\binom{6}{2} = \frac{6 \cdot 5}{2} = \frac{30}{2}$$

$\boxed{15}$

---

11. Expand $(x^2 + 3)^3$.
Solution: Treat $A = x^2$, so each power of A doubles the exponent: $A^3 = x^6$, $A^2 = x^4$, $A^1 = x^2$. Row 3 is $1, 3, 3, 1$.

$$(x^2 + 3)^3 = x^6 + 3x^4(3) + 3x^2(9) + 27$$

$\boxed{x^6 + 9x^4 + 27x^2 + 27}$

---

12. (a) Expand $(1 + x)^4$. (b) Use it to evaluate 3^4. (c) Find n with coefficient sum 512, and explain.
Solution (a): Row 4 is $1, 4, 6, 4, 1$, and every power of 1 is 1: $\boxed{x^4 + 4x^3 + 6x^2 + 4x + 1}$
Solution (b): $3^4 = (1 + 2)^4$, so substitute $x = 2$:

$$16 + 4(8) + 6(4) + 4(2) + 1 = 16 + 32 + 24 + 8 + 1$$

$\boxed{81}$

Solution (c): Substituting $x = 1$ makes every power of x equal to 1, so the expansion collapses to the sum of its coefficients — and the left side becomes 2^n.

$$2^n = 512 = 2^9$$

$$\boxed{n = 9}$$

The written half of (c) — instructor-judged. A correct response says that substituting 1 for the letter turns every power into 1, so each term is left as its own coefficient and the total of the expansion is the total of the coefficients; since $1 + 1 = 2$, that total is 2^n. Full credit connects the substitution to the collapse of the powers. Writing $2^n = 512$ and solving, with no account of *why* the coefficient sum is a power of 2, answers only half the question.
